NoSQL

NoSQL (Not Only SQL) is a type of database management system that provides a flexible approach to storing and retrieving data.

Unlike traditional **relational databases (RDBMS)** that use structured tables and SQL (Structured Query Language), NoSQL databases are designed for **scalability, high performance, and handling unstructured or semi-structured data**.

NoSQL databases are commonly used in **big data, real-time web applications, IoT systems, and cloud-based services** due to their ability to handle **large-scale distributed data** efficiently.

Features of NoSQL Databases

1. **Schema Flexibility**
   - Unlike relational databases, NoSQL databases do not require a fixed schema.
   - They allow dynamic changes in data structure without affecting existing records.
   - Example: MongoDB allows adding new fields to documents without altering the existing structure.
2. **Scalability**
   - NoSQL databases support **horizontal scaling** (adding more servers) instead of **vertical scaling** (upgrading hardware).
   - This makes them ideal for handling high-traffic applications.
   - Example: Facebook and Twitter use NoSQL databases for managing user-generated content.

Features of NoSQL Databases

3. **High Performance**
   - NoSQL databases use **distributed architecture** to ensure fast read/write operations.
   - They support caching mechanisms and optimized queries for better performance.
4. **Handling Unstructured and Semi-structured Data**
   - NoSQL databases efficiently store and manage **JSON, XML, key-value pairs, graphs, and documents**.
   - Example: IoT applications store real-time sensor data using NoSQL databases like Cassandra.
5. **Support for Big Data and Real-time Processing**
   - NoSQL databases are designed to handle **large volumes of data** and real-time analytics.
   - Example: Apache Cassandra is used by Netflix to store and analyze streaming data.

Example of NoSQL Databases

NoSQL Database	Type	Use Case
MongoDB	Document Store	Storing JSON-like flexible documents for web apps
Redis	Key-Value Store	Caching and session management
Cassandra	Column-Family Store	Distributed and scalable databases for big data applications
Neo4j	Graph Database	Social networks, fraud detection, and recommendation engines
Amazon DynamoDB	Key-Value & Document Store	Serverless applications and IoT data storage

Types of NoSQL Databases

NoSQL databases are classified into four main types based on their data storage and retrieval models. These are:

1. **Key-Value Store**
2. **Document Store**
3. **Column-Family Store**
4. **Graph Database**

Types of NoSQL Databases

Key Value Pair Based

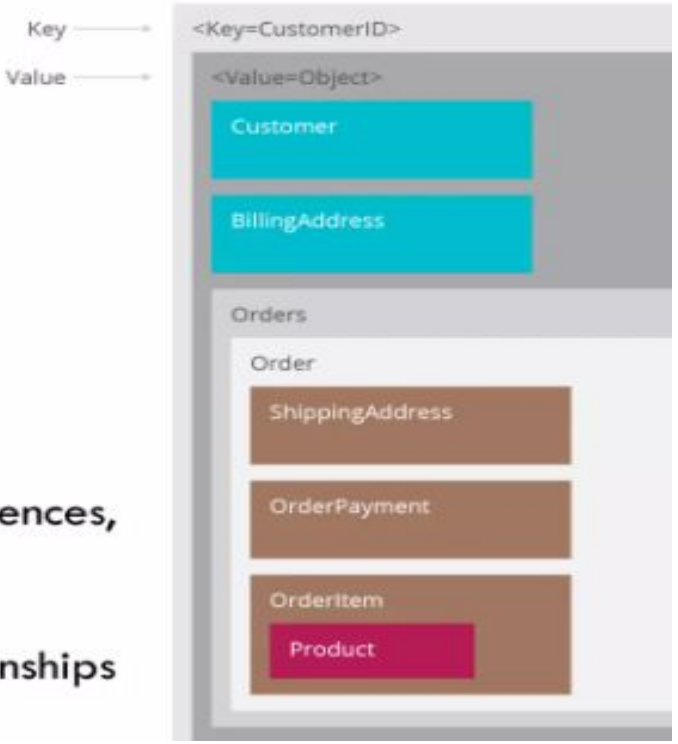
8

- Designed for processing dictionary. Dictionaries contain a collection of records having fields containing data.
- Records are stored and retrieved using a key that uniquely identifies the record, and is used to quickly find the data within the database.

Example: CouchDB, Oracle NoSQL Database, Riak etc.

We use it for storing session information, user profiles, preferences, shopping cart data.

We would avoid it when we need to query data having relationships between entities.



Types of NoSQL Databases

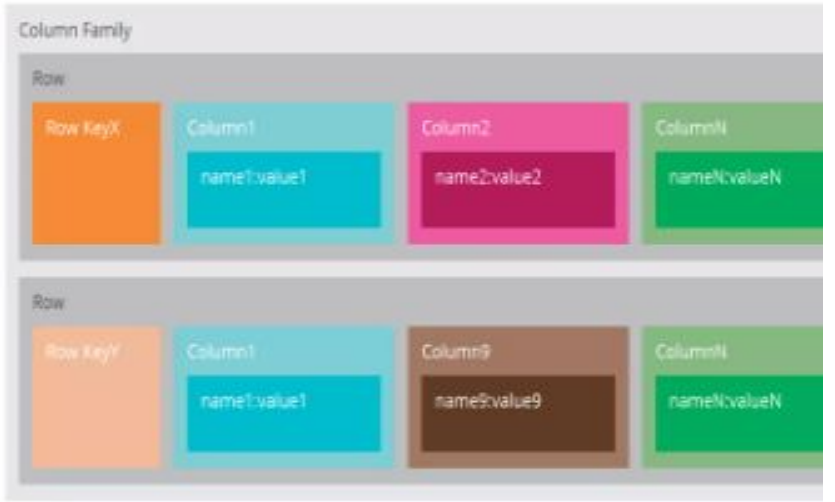
Column based

9

It store data as Column families containing rows that have many columns associated with a row key. Each row can have different columns.

Column families are groups of related data that is accessed together.

Example: Cassandra, HBase, Hypertable, and Amazon DynamoDB.



We use it for content management systems, blogging platforms, log aggregation.

We would avoid it for systems that are in early development, changing query patterns.

Types of NoSQL Databases

Document Based

10

The database stores and retrieves documents. It stores documents in the value part of the key-value store.

Self- describing, hierarchical tree data structures consisting of maps, collections, and scalar values.

Example: Lotus Notes, MongoDB, Couch DB, Orient DB, Raven DB.



We use it for content management systems, blogging platforms, web analytics, real-time analytics, e- commerce applications.

We would avoid it for systems that need complex transactions spanning multiple operations or queries against varying aggregate structures.

Types of NoSQL Databases

Graph Based

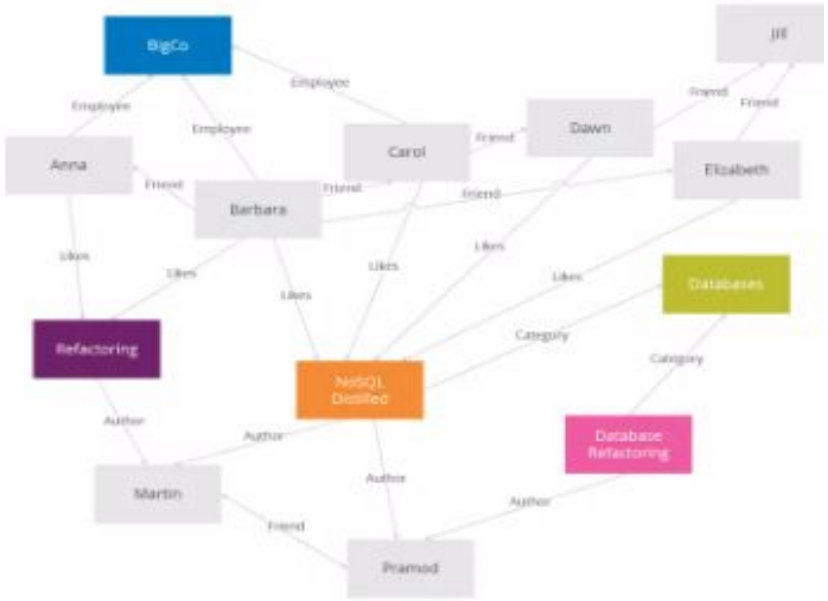
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Store entities and relationships between these entities as nodes and edges of a graph respectively. Entities have properties.

Traversing the relationships is very fast as relationship between nodes is not calculated at query time but is actually persisted as a relationship.

Example: Neo4J, Infinite Graph, OrientDB, FlockDB.

It is well suited for connected data, such as social networks, spatial data, routing information for goods and supply.



Types of NoSQL Databases

CAP Theorem

12

- According to Eric Brewer a distributed system has 3 properties :
  - Consistency
  - Availability
  - Partitions
- We can have at most two of these three properties for any shared-data system
- To scale out, we have to partition. It leaves a choice between consistency and availability. *(In almost all cases, we would choose availability over consistency)*
- Everyone who builds big applications builds them on CAP : Google, Yahoo, Facebook, Amazon, eBay, etc.

Advantages of NoSQL

- Cheap and easy to implement (open source)
- Data are replicated to multiple nodes (therefore identical and fault-tolerant) and can be partitioned
  - When data is written, the latest version is on at least one node and then replicated to other nodes
  - No single point of failure
- Easy to distribute
- Don't require a schema

What is not provided by NoSQL

- Joins
- Group by
- ACID transactions
- SQL
- Integration with applications that are based on SQL

Where to use NoSQL

- NoSQL Data storage systems makes sense for applications that process **very large semi-structured data** –like Log Analysis, Social Networking Feeds, Time-based data.
- To improve programmer productivity by using a database that better matches an application's needs.
- To improve data access performance via some combination of handling larger data volumes, reducing latency, and improving throughput.

Feature	SQL (Relational Database)	NoSQL (Non-Relational Database)
Data Structure	Uses tables with rows and columns (structured data)	Stores data in key-value, document, column-family, or graph formats (unstructured/semi-structured data)
Schema	Fixed schema (predefined structure)	Flexible schema (can store different types of data in a single database)
Scalability	Vertically scalable (requires upgrading hardware)	Horizontally scalable (can add more servers to handle load)
Query Language	Uses SQL for queries (structured query language)	Uses different query mechanisms (JSON, key-value lookups, graph traversal, etc.)
ACID Compliance	Strong ACID (Atomicity, Consistency, Isolation, Durability) properties	Often follows BASE (Basically Available, Soft-state, Eventually consistent) for high availability
Performance	Can be slower for large-scale read/write operations	Optimized for high-speed read/write and big data processing
Best Used For	Structured data, transactions, financial apps, ERP, CRM	Big data, IoT, real-time analytics, social media, distributed apps
Examples	MySQL, PostgreSQL, Oracle, SQL Server	MongoDB, Cassandra, Redis, Neo4j, DynamoDB